

A space-frequency joint detection and tracking method for line-spectrum components of underwater acoustic signals

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ABSTRACT

For the passive detection of azimuth and frequency of azimuths and frequencies of line-spectrum signals radiated from underwater weak targets, a space-frequency joint detection and tracking method based on track-before-detect (TBD) technology is proposed. This method works directly with the time sequence of frequency-azimuth (FRAZ) spectra to estimate the frequency and azimuth of line-spectrum signals. And the change of the frequency and azimuth states of a line-spectrum signal is modeled as a first-order hidden Markov process. The distribution of the FRAZ spectra of linear array received signal are derived in this paper to set the parameters of Hidden Markov Models (HMM). The FRAZ spectra at different times is joint process using Viterbi algorithm to obtain the maximum a posteriori estimation of the azimuth and frequency of line-spectrum signal. A dynamic searching method based on region segmentation and merging similar line-spectrum is designed to solve the problem of large computation of Viterbi algorithm in the detection of line-spectrum. The proposed method can make full use of the correlation between the target's information in the multi-frame FRAZ spectra to distinguish the target from the background noise. The processing results of simulation data and actual data show that the method has excellent line-spectrum detection performance at low SNR and can adapt to the complex situation such as there are multiple targets and multiple line-spectrum signals.

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1. Introduction

Sonar array is usually used in the detection of long-distance target signal to acquire the spatial gain and the azimuth estimation. And the low frequency line-spectrum signal in ship radiated noise has the characteristics of high intensity, good stability and low transmission loss [1–3] and thus is of great significance to detect low noise and quiet underwater targets. However, the performance of traditional target detection methods is limited because of the development of suppression technology on vibration and noise. Therefore, it is of great significance to study how to use the line-spectrum signal to detect the underwater target more effectively.

At present, the estimation of the frequency and azimuth of the target radiation signal is usually independent. Among the methods to determine the azimuth of the targets, the conventional energy detection (ED) is easy to submerge the target radiation signal in the wide-band background noise because the information of line-spectrum signals is not fully used [4]. In the FRAZ spectra, the amplitude at line-spectrum's azimuth and frequency is more stable compared with that at noise. Using this feature, the method based

on frequency variance weighting detection (FVWD) [5] calculates the peak frequency variance of different beam directions to weighted the power-bearing spectrum obtained by traditional beamforming. And the method based on azimuth stability detection (ASD) [6] detect line-spectrum targets by using the difference of azimuth stability between noise and line-spectrum. However, the performance of method in [5] and [6] will be seriously decrease when SNR is low or there are multiple line-spectrum signals. A method based on Hidden Markov Model (HMM) is proposed to extract target azimuth trajectory in [7], which reduces the root mean square error of estimated trajectory and maintains lower false alarm probability. But this method is processed on the bearings vs. time recording composed of multiple frames of power-bearing spectrum, which does not make full use of the frequency information of line-spectrum signals. The frequencies of line-spectrum signals are usually determined according to the beam-forming results in the estimated target azimuth. The most classical line-spectrum detection method is based on the periodogram and the Welch power spectrum estimation. Firstly, the power spectrum of the received signal is estimated by segmented DFT processing. Then the background noise is equalized by the methods of α filtering, the fitting method based on empirical mode decomposition (EMD) or the fitting method based on regression parameters

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[8–10]. Finally, the detection result of line-spectrum is obtained by amplitude threshold judgment of power spectra. In addition, because of the stability of line-spectrum in time dimension, the frequency tracking can also be realized in lofargram. The methods of line-spectrum detection in lofargram can be divided into three kinds: statistical model based method, image processing based method and neural network based method [11]. Among them, the HMM based line-spectrum detection method [12] has excellent performance under the condition of low SNR. But the computational burden of this method is very high. Reference [13] greatly reduces the complexity of HMM based detection method, and improves the adaptability of the algorithm in the presence of multiple time-varying line spectrum signals by using image stitching theory and statistical model method.

Although the above methods can achieve excellent detection performance under specific conditions, the frequency information of the targets is not sufficiently used when detecting the azimuths. And the premise of good performance when detecting the targets' frequencies is to know the azimuths. A space-frequency joint detection and tracking (SFJDT) method of detecting line-spectrum signals based on TBD is proposed to solve the above problems. This paper models the change of the azimuth and frequency of the target radiated line-spectrum signal as a first-order Markov process, and directly processes the time sequence of FRAZ spectra. The methods of dynamic search and similar line-spectrum merging in different regions are designed to solve the problem of large computation in the detection of line-spectrum, which make the proposed algorithm meet the real-time processing condition on personal PC platform. In the proposed method, the FRAZ spectra are divided into sub-regions and the sub-regions which obviously does not exist target are rejected by high false alarm detection. The maximum a posteriori (MAP) estimation of the azimuths and frequency of the line-spectrum signal in each sub-region is obtained using Viterbi algorithm in the next step. Finally, line-spectrum merging is used to process the line-spectrum detected in different sub-regions to get complete information of the line-spectrum signals. The proposed method can determine the azimuth and frequency information of the target line-spectrum at the same time. It can adapt to the complex situation of multiple line-spectrum signals in multiple azimuths, and has good detection performance under the condition of low SNR.

2. System model

2.1. Array signal reception model

We assume that H line-spectrum signals from different targets in space are incident to Q -element uniform line array in the form of far-field plane wave. The environmental noise is isotropic, meanwhile the target signal and the environmental noise are independent at the receiving end of the signal. Fig. 1 shows the array receiving model of the h -th line-spectrum signal.

In Fig. 1, the initial phase of the line-spectrum signal follows the uniform distribution, which is ϕ_h . The angle between the target and the normal direction of the array is θ_h , $h = 1, 2, \dots, H$. When the sampling rate is f_s and array element 1 is the reference element, the received signal of the q -th element can be described as:

$$x_q(l) = g_q(l) + \sum_{h=1}^H A_h \cos(2\pi f_h l / f_s + q\phi_h + \phi_h) \quad (1)$$

In Eq. (1), f_h is the signal frequency, A_h is the signal amplitude, g_q is the broadband noise with power δ^2 , and l is the discrete-time sequence number. The phase difference of the line-spectrum signal received by the adjacent array element $\phi_h = 2\pi f_h d \sin \theta_h / C$, d is the

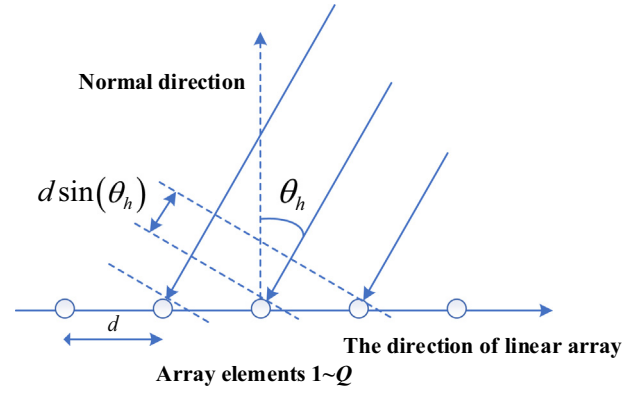


Fig. 1. Signal reception diagram for uniform linear array.

distance between the array elements, and C is the sound velocity in water. Suppose $f_h \in S_f$, $\theta_h \in S_\theta$, where S_f is the frequency set of line-spectrum signal, S_θ is the azimuth set of line-spectrum. In the actual passive detection of underwater target, S_f and S_θ are the results we want to estimate.

The signals received by Q array elements at time l are defined as

$$X(l) = [x_0(l) \cdots x_q(l) \cdots x_{Q-1}(l)]^T \quad (2)$$

The K -segment sampling data is taken out from the received data, and each segment contains L sampling points. Then beamforming is carried out on each sampling data. In order to facilitate the analysis, this paper adopts the frequency domain beamforming method. The FRAZ spectra of k -th segment of data are as follows

$$P_k(\theta, f) = \frac{1}{QL} \left| \sum_{q=0}^{Q-1} \sum_{l=0}^{L-1} x_q(l + kL) e^{-j2\pi l f / f_s} e^{-j2\pi q d \sin(\theta) / C} \right|^2 \quad (3)$$

In Eq. (3), f is the possible frequency of the line-spectrum, and θ is the possible azimuth of target. Assume that the set of all possible frequencies in the effective frequency band is defined as $\mathbf{F}^N$, and the set of all possible beam directions is Θ^M , where M is the total number of directions and N is the total number of frequency points. Fig. 2 shows the change of the azimuth and frequency state of a single line-spectrum signal in the FRAZ spectra.

The boxes in Fig. 2 represent the time sequence of the FRAZ spectra, and the coordinate trajectory is illustrated by a line. The key problem of this paper is to detect the unknown line-spectrum in time sequence of the FRAZ spectra and to estimate its azimuth and frequency.

2.2. HMM parameters setting

The change of the frequency and azimuth of the line-spectrum signal is modeled as a first-order hidden Markov process in this paper. Therefore, it is necessary to determine the parameter set of HMM: $\{Q, V, A, B, \Pi\}$ using the time sequence of the FRAZ spectra. In the parameter set, Q is the set of all finite states; V is the finite measurement set; A is the transition probability matrix; B is the measurement probability matrix; Π is the initial state probability vector of the Markov chain. Corresponding to the time sequence of the FRAZ spectra, the set of all finite states $Q = \{[f \ \theta]^T, \text{s.t. } f \in \mathbf{F}^N, \theta \in \Theta^M\}$. The state at time k is defined as

$$\mathbf{q}_k = [f_k \ \theta_k]^T = [q_{k,1} \ q_{k,2}]^T \quad (4)$$

It obeys the following state equation

$$\mathbf{q}_k = \mathbf{q}_{k-1} + \boldsymbol{\eta}_k \quad (5)$$

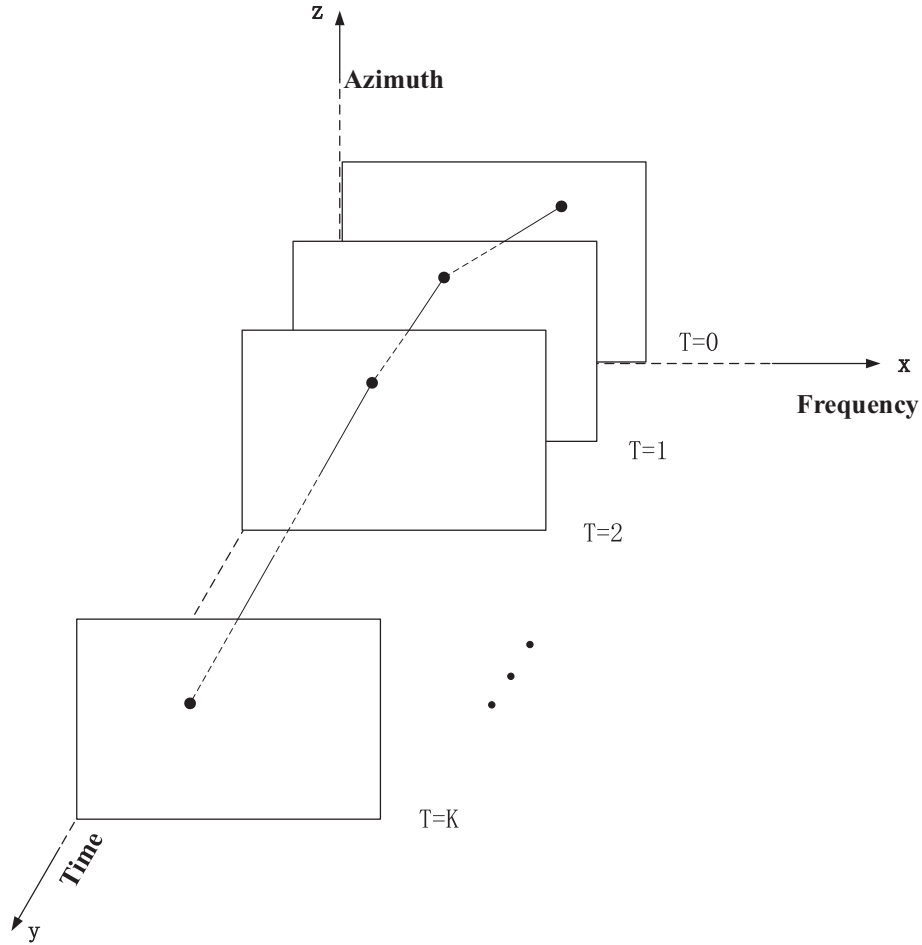


Fig. 2. Schematic showing a line-spectrum through the time history of two-dimensional FRAZ spectra.

The term f_k and θ_k represent the frequency state and azimuth state of the line-spectrum and $f_k \in \mathbf{F}^N$, $\theta_k \in \Theta^M$. $\boldsymbol{\eta}_k$ represents the plan noise; it is 0-mean and of covariance matrix $\mathbf{R} = \begin{bmatrix} \delta_f^2 & \\ & \delta_\theta^2 \end{bmatrix}$.

On the other side, the measurement set are given by the FRAZ spectra

$$\mathbf{z}_k = (\mathbf{z}_{k,0,0}, \mathbf{z}_{k,1,0}, \dots, \mathbf{z}_{k,N-1,0}, \mathbf{z}_{k,0,1}, \dots, \mathbf{z}_{k,N-1,M-1}) \quad (6)$$

Where

$$\mathbf{z}_{k,i,r} = \sqrt{P_k(r,i)/LQ}, i \in \mathbf{F}^N, r \in \mathbf{B}^M \quad (7)$$

The finite measurement set $\mathbf{V}$ is denoted as $\{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_K\}$.

As for the transition probability matrix $\mathbf{A}$, whose element is defined as

$$\begin{aligned} a_{j,s}(i,r) &= P_r(\mathbf{q}_k = [i \ r]^T | \mathbf{q}_{k-1} = [j \ s]^T) \\ &= \frac{\Omega}{2\pi\sqrt{\det \mathbf{R}}} \exp\left(-\frac{1}{2} \|\mathbf{q}_k - \mathbf{q}_{k-1}\|^2 \mathbf{R}^{-1}\right) \end{aligned} \quad (8)$$

with $i, j \in \mathbf{F}^N$, $r, s \in \mathbf{B}^M$. It means the probability of state $\mathbf{q}_{k-1}$ transferring to state $\mathbf{q}_k$. And Ω is a scaling factor making $a_{j,s}(i,r)$ satisfies equation

$$\sum_{i=1}^N \sum_{r=1}^M a_{j,s}(i,r) = 1 \quad (9)$$

For the elements in measurement probability matrix $\mathbf{B}$, we have the following relation with our measurement model

$$b_{i,r}(\mathbf{z}_k) = P_r(\mathbf{z}_k | \mathbf{q}_k = [i \ r]^T) \quad (10)$$

$b_{i,r}(\mathbf{z}_k)$ represents the probability that state $\mathbf{q}_k = [i \ r]^T$ when the measurement is $\mathbf{z}_k$.

When there is only one target, according to Eq. (1), the received signal of q -th array element can be denoted by

$$x_q(l) = A \cos(2\pi f_0 l / f_s + q\varphi + \phi) + g_q(l) \quad (11)$$

Where f_0 is the signal frequency, the phase difference $\varphi = 2\pi f d \sin \theta_0 / C$. θ_0 is the angle between the target and the normal direction of the array.

And it is well known that a narrow band Gaussian noise with zero mean and variance σ_c^2 plus a sine wave $A_c \cos(2\pi f_c t + \xi)$ with f_c being in this band and ξ being a random variable uniformly distributed on $[0, 2\pi]$ will appear as

$$e(t) \cos(2\pi f_c t + \theta_t) \quad (12)$$

The probability density function of the envelope $r(t)$ [14] is.

$$P_1(e) = \begin{cases} \frac{e}{\sigma_c^2} \exp\left(-\frac{e^2 + A_c^2}{2\sigma_c^2}\right) I_0\left(\frac{A_c e}{\sigma_c^2}\right), & e \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad (13)$$

And if there is only a narrow band Gaussian noise with zero mean and variance σ_c^2 , the probability density function of $e(t)$ will be

$$P_0(e) = \begin{cases} \frac{e}{\sigma_c^2} \exp\left(-\frac{e^2}{2\sigma_c^2}\right), & e \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad (14)$$

For our measurements, each frequency-azimuth cell can be seen as a narrow band of frequency. The standard deviation of noise in a cell is

$$\sigma_c = \sqrt{2\Delta f/f_s} \sigma = \sqrt{2/L} \sigma \quad (15)$$

Among all measurements at time k , $\mathbf{z}_{k,f_0,\theta_0}$ at the frequency and azimuth of the target line-spectrum is the maximum value. And $\mathbf{z}_{k,f_0,r}$, s.t. $r \neq \theta_0$ is related to the directivity function of linear array. Substituting Eqs. (3), (7), (13), (14) and (15), we hence have the likelihood of $\mathbf{z}_{k,i,r}$

$$\begin{cases} P_r(\mathbf{z}_{k,i,r}|\mathbf{q}_k = [j \ s]^T, i \neq j) = \frac{2LQ\mathbf{z}_{k,i,r}}{\sigma_c^2} \exp\left(-\frac{LQ\mathbf{z}_{k,i,r}^2}{\sigma_c^2}\right) \\ P_r(\mathbf{z}_{k,i,r}|\mathbf{q}_k = [j \ s]^T, i = j) = \frac{2LQ\mathbf{z}_{k,i,r}}{\sigma_c^2} \exp\left(-\frac{LQ\mathbf{z}_{k,i,r}^2 + LQD_s^2(j,r)A^2}{\sigma_c^2}\right) \# \\ *I_0\left(\frac{2D_s(j,r)ALQ\mathbf{z}_{k,i,r}}{\sigma_c^2}\right) \end{cases} \quad (16)$$

Where, $D_s(j, r)$ is the directivity function of linear array, which is denoted by

$$D_s(j, r) = \left| \frac{\sin[N\pi j d(\sin(r) - \sin(s))/f_s C]}{N \sin[\pi j d(\sin(r) - \sin(s))/f_s C]} \right| \quad (17)$$

When assuming that the $\mathbf{z}_{k,i,r}$'s are conditionally independent, we get

$$\begin{aligned} P_r(\mathbf{z}_k|\mathbf{q}_k = [j \ s]^T) &= \frac{(2LQ)^{MN} \prod_{r=0}^{M-1} \prod_{i=0}^{N-1} \mathbf{z}_{k,i,r}}{\sigma_c^{2MN}} \exp\left(-\frac{LQ \sum_{r=0}^{M-1} \sum_{i=0}^{N-1} \mathbf{z}_{k,i,r}^2}{\sigma_c^2}\right) \\ &* \exp\left(-\frac{LQA^2 \sum_{r=0}^{M-1} D_s^2(j,r)}{\sigma_c^2}\right) \prod_{r=0}^{M-1} I_0\left(\frac{2D_s(j,r)ALQ\mathbf{z}_{k,i,r}}{\sigma_c^2}\right) \end{aligned} \quad (18)$$

According to the above equation and Eq. (10), we get the value of $b_{i,r}(\mathbf{z}_k)$. From the principle of the Viterbi algorithm, we can know that the Viterbi algorithm will give us the same result if we multiply all $b_{i,r}(\mathbf{z}_k)$ by the same scale factor. Therefore, the element in measurement probability matrix $\mathbf{B}$ can be defined as follows

$$b_{i,r}^s(\mathbf{z}_k) = \frac{b_{i,r}(\mathbf{z}_k)}{\sum_{s=0}^{M-1} \sum_{j=0}^{N-1} b_{j,s}(\mathbf{z}_k)} \quad (19)$$

However, there is usually no prior knowledge in the process of passive sonar target detection, so we need to approximate the above equation. For the directivity function of linear array, the energy is mainly concentrated in the main lobe. As a result, the value of $\exp\left(-\frac{LQA^2 \sum_{r=0}^{M-1} D_s^2(j,r)}{\sigma_c^2}\right)$ in different azimuth is very close.

In addition, the value of $\log(P_r(\mathbf{z}_{k,i,r}|\mathbf{q}_k = [j \ s]^T, j = i))$ changes linearly with the signal amplitude when the background noise is Gaussian white noise [15]. And the measurement likelihood can be replaced by amplitude of power spectra under the condition of low SNR [16]. Using the same approximation method, the element in measurement probability matrix $\mathbf{B}$ is defined as

$$b_{i,r}^s(\mathbf{z}_k) = \frac{\hat{P}_r(\mathbf{z}_k|\mathbf{q}_k = [i \ r]^T)}{\sum_{s=0}^{M-1} \sum_{j=0}^{N-1} \hat{P}_r(\mathbf{z}_k|\mathbf{q}_k = [j \ s]^T)} \quad (20)$$

Where,

$$\hat{P}_r(\mathbf{z}_k|\mathbf{q}_k = [j \ s]^T) = \sum_{r=0}^{M-1} D_s^2(j, r) P_k(r, j) \quad (21)$$

And $D_s^2(j, r)$ can be calculated in advance to reduce the amount of calculation in the actual processing.

When there are multiple targets, the line-spectrum signals of the targets are analyzed one by one. The element in $\mathbf{B}$ is denoted as $b_{i,r}^{s,h}(\mathbf{z}_k)$, which represents the probability that the state of the h -th target is $[j \ s]^T$ and the measurement is $\mathbf{z}_k$. As H line-spectrum signals are mutually independent, we can write

$$b_{i,r}^{s,h}(\mathbf{z}_k) \propto \frac{P_r(\mathbf{z}_k|\mathbf{q}_k^h = [i \ r]^T)}{\sum_{s=0}^{M-1} \sum_{j=0}^{N-1} P_r(\mathbf{z}_k|\mathbf{q}_k^h = [j \ s]^T)} \quad (22)$$

And

$$P_r(\mathbf{z}_k|\mathbf{q}_k^h = [j \ s]^T) = \prod_{r=0}^{M-1} I_0\left(\frac{2D_s(j, r)\hat{A}^j LQ\mathbf{z}_{k,j,r}}{\sigma_c^2}\right) \quad (23)$$

where $\hat{A}^j$ is the amplitude's mean contribution of the H targets in the frequency j . Using the same method for approximation, the quantity $b_{i,r}^{s,h}(\mathbf{z}_k)$ can also be approximated by Eq. (20).

There is usually no prior information of line-spectrum parameters in the process of passive target detection. We hence assume that the initial state probability vector Π is uniformly distributed in the frequency and azimuth dimensions, and define $\Pi(i, r) = \frac{1}{MN}$.

3. Space-Frequency joint detection and tracking algorithm

The detection and tracking of line-spectrum in the time sequence of the FRAZ spectra can be divided into four steps: data preprocessing, region segmentation and screening, line-spectrum detection and line-spectrum merging. And the framework of the algorithm is shown in Fig. 3.

3.1. Data preprocessing

In order to accurately and reliably estimate the frequency and azimuth of line-spectrum signals, it is necessary to perform background equalization on the received signal to reduce the interference of noise on line-spectrum detection. The direction of the target is determined according to the amplitude difference of the FRAZ spectra in the azimuth dimension. Therefore, the background equalization needs to keep the relative magnitude of line-spectrum in different beam directions. In this paper, we use the fitting method based on regression parameters [10] to estimate the continuous spectrum in FFT results of each array element before beamforming to achieve background equalization.

3.2. Region segmentation and screening

The calculation amount of HMM based dynamic programming method is square to the number of states when detecting line-spectrum in the time sequence of the FRAZ spectra. The calculation will extremely huge if the FRAZ spectra of all directions and frequencies is processed at the same time. Therefore, in the proposed method, the FRAZ spectra is segmented into sub-regions before line-spectrum detection to reduce the number of states in each iteration. In addition, the FRAZ spectra contains a large number of azimuth and frequency states, while the frequency and azimuth of the target line-spectrum only account for a small part, so there is no line-spectrum in most of the sub-regions. In order to reduce the amount of calculation, the existence of line-spectrum in the sub-region can be preliminarily judged by high false alarm detection. The next frequency and azimuth estimation of the line-spectrum is only carried out for the sub-regions that meet the judgment conditions.

The region segmentation and screening results of the time sequence of the FRAZ spectra are shown in Fig. 4. There is a certain

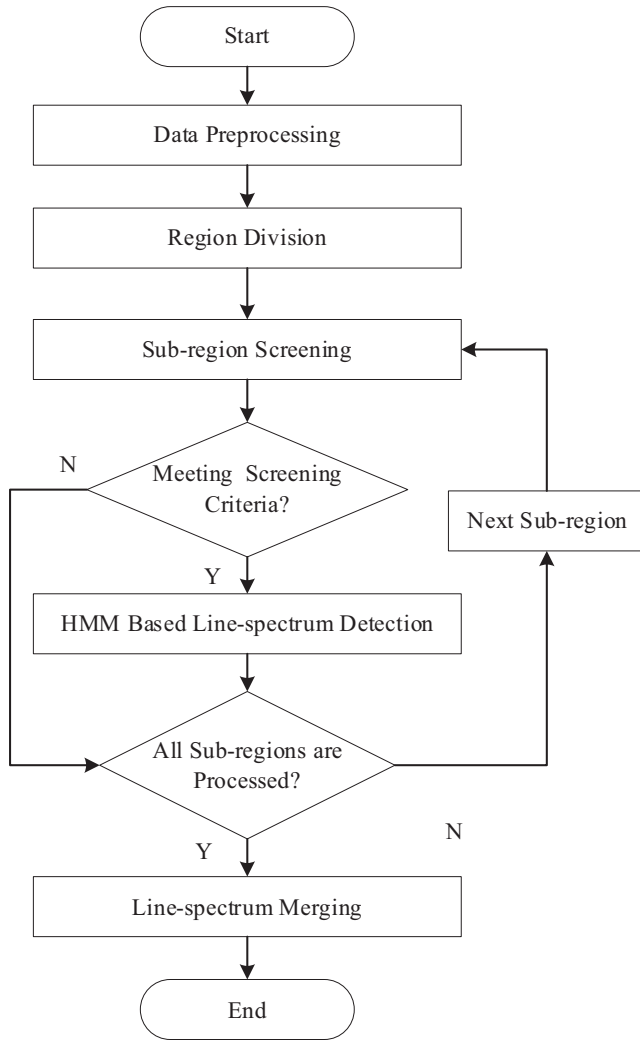


Fig. 3. Schematic showing the framework of proposed algorithm.

degree of overlap between adjacent region in frequency dimension, azimuth dimension and time dimension. And the size of the overlapping parts and the size of the regions are both determined according to the actual needs.

In Fig. 4, the time dimension is divided into two parts: $0 \sim 3K/4$ and $K/4 \sim K$, and the length of overlap part is $K/2$. Azimuth dimension and frequency dimension are also divided into two parts with a certain degree of overlap.

In the process of region screening, the sub-region to be judged is first transformed into lofargram in multiple beam directions. Then, the lofargram screening method mentioned in reference [13] is used to judge the multiple lofargrams one by one. If there is lofargram in a sub-region that satisfies the judgment condition, the sub-region will be marked as a region to be processed; Otherwise, the sub-region will be eliminated and no line-spectrum detection will be performed on it.

3.3. Line-spectrum detection

After region segmentation and screening, the marked azimuth-frequency spectra at different times is then joint process using Viterbi algorithm to sequentially obtain the maximum a posteriori estimation of the azimuth and frequency of each line-spectrum signal. The detection process is shown in Fig. 5.

According to Fig. 5, the HMM parameters $\{A, B, \Pi\}$ need to be calculated first according to Section 2.2. And it is necessary to define two variables for recursion when estimating line-spectrum information using Viterbi algorithm. The maximum probability of the time sequence of states $\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_k$ with state $\mathbf{q}_k = [i \ r]^T$ is defined by

$$\delta_k(i, r) = \max_{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_{k-1}} P(\mathbf{q}_k = [i \ r]^T, \mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_{k-1}, \mathbf{Z}_k, \mathbf{Z}_{k-1}, \dots, \mathbf{Z}_1) \quad (24)$$

with $i = 1, 2, \dots, N, r = 1, 2, \dots, M$. The value of $\mathbf{q}_{k-1}$ in the sequence $\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_k$ with the maximum probability of all the time sequences of states is defined by

$$\eta_k(i, r) = \arg \max_{[j \ s]^T} [\delta_{k-1}(j, s) a_{j,s}(i, r)] \quad (25)$$

$$1 \leq j \leq N, 1 \leq s \leq M$$

Before performing line-spectrum estimation, it is supposed to preset the maximum number of line-spectrum signals G in the marked FRAZ spectra according to prior information. From the principle of the Viterbi algorithm, we can know that the Viterbi algorithm will give an optimal search sequence. In this paper, with the iteration step by step, the probability $\delta_k(i, r)$ of the optimal state will gradually increase. In other words, when there are multiple targets, this phenomenon will lead to the suppression of other components except the strongest line spectrum. The other components here include not only the background noise, but also the rest of the line-spectrum. Therefore, we empirical assumed that there is only one line-spectrum in the marked sub-region before each line-spectrum detection. The line-spectrum with the maximum a posteriori probability is taken from the FRAZ spectra as the result of line-spectrum detection this time. When tracking the next line-spectrum signal, the detected line-spectrum needs to be deleted from the marked time sequence of FRAZ spectra. The line-spectrum is estimated in the new generated sequence until all possible G line-spectrum signals are detected. The processing method is described in detail below.

1. Initialize the variables $\delta_k(i, r)$ and $\eta_k(i, r)$ corresponding to all states as follows

$$\begin{aligned} \delta_1(i, r) &= \Pi(i, r) b_{i,r}^s(\mathbf{z}_1) \\ \eta_1(i, r) &= [0 \ 0]^T \end{aligned} \quad (26)$$

2. For each state $[i \ r]^T$ at time $k = 2, 3, \dots, K$, $\delta_k(i, r)$ is calculated and its previous state $\eta_k(i, r)$ is stored using the recursions

$$\begin{aligned} \delta_k(i, r) &= \max_{[j \ s]^T} [\delta_{k-1}(j, s) a_{j,s}(i, r)] b_{i,r}^s(\mathbf{z}_k) \\ \eta_k(i, r) &= \arg \max_{[j \ s]^T} [\delta_{k-1}(j, s) a_{j,s}(i, r)] \end{aligned} \quad (27)$$

3. After the recursion is completed, the state estimate at time K is found by maximizing $\delta_K(i, r)$, and the estimated time sequence of states is obtained by backtracking using the backward state stored in η_k . The step can be denoted as

$$\begin{aligned} \hat{\mathbf{q}}_K &= \arg \max_{[i \ r]^T} [\delta_K(i, r)] \\ \hat{\mathbf{q}}_{k-1} &= \eta_k(\hat{\mathbf{q}}_k) \end{aligned} \quad (28)$$

Where, $k = K, \dots, 2$. The estimated result of line-spectrum is $(\hat{\mathbf{q}}_1, \hat{\mathbf{q}}_2, \dots, \hat{\mathbf{q}}_K)$ this time.

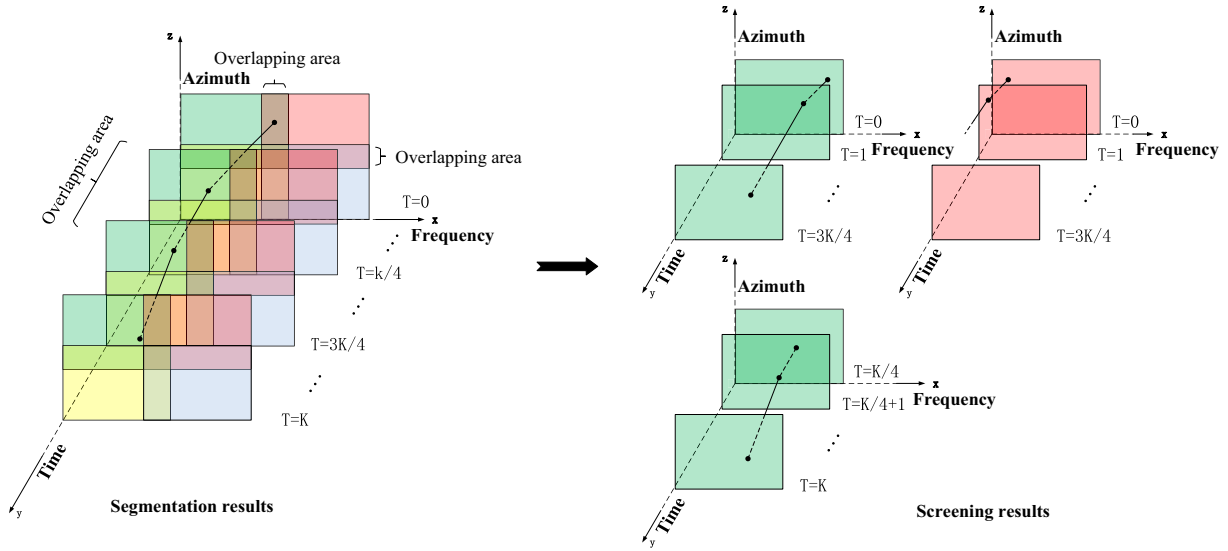


Fig. 4. Schematic showing region segmentation and screening results.

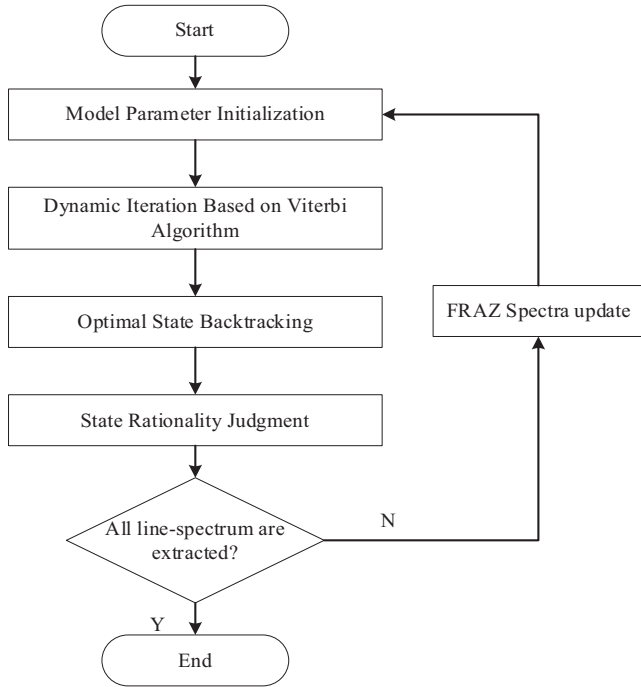


Fig. 5. Process of line-spectrum detection.

4. The amplitude-based decision method is used to determine the rationality of the estimated line-spectrum. According to the Viterbi algorithm, an optimal estimated result of line-spectrum from time 1 to time K can be obtained after step 1–3. However, the elements in the estimated time sequence of states are not necessarily generated by the actual line-spectrum signals. It is necessary to make a second decision state by state on all states in the estimated time sequence. The amplitude of line-spectrum in the FRAZ spectra is an abnormal value in frequency dimension and azimuth dimension. As a result, we can advantageously make a judgment by the equation

$$\begin{cases} \hat{q}_{k,1} \in S_f, \hat{q}_{k,2} \in S_{\theta} P_k(\hat{q}_{k,2}, \hat{q}_{k,1}) \geq Th \\ Falsealarm, P_k(\hat{q}_{k,2}, \hat{q}_{k,1}) \leq Th \end{cases} \quad (29)$$

And the threshold $Th = \alpha \bar{P}_k(\hat{q}_{k,2}, \hat{q}_{k,1})$, where $\bar{P}_k(\hat{q}_{k,2}, \hat{q}_{k,1})$ is the mean value of the amplitude of the corresponding neighborhood of detected line-spectrum in FRAZ spectra. And the value of α is set according to the actual SNR.

5. Remove the information related to the detected line-spectrum in the marked time sequence of FRAZ spectra. According to the directivity function of linear array, pseudo-peak will exist in several beam directions near the target azimuth in FRAZ spectra. Therefore, it is necessary to eliminate the pseudo-peak generated by detected line-spectrum in other azimuth of FRAZ spectra. We can get the theoretical pseudo-peak amplitudes in other azimuth by using the information of the detected line-spectrum this time through Eq. (17). Then the calculated pseudo-peak amplitudes are subtracted from the corresponding position of FRAZ spectra to obtain a new region for the next line-spectrum detection.

4. Repeat steps 1–5 until the detection of preset G line-spectrum signals is completed.

4.1. Line-spectrum merging

The distance matrix of line-spectrum in different regions is calculate in the process of line-spectrum merging [13]. The distance matrix of $i-1$ -th region and i -th region is defined as follows:

$$J_{i-1,i} = \begin{bmatrix} \Delta_{11} & \Delta_{12} & \cdots & \Delta_{1G_i} \\ \Delta_{21} & \Delta_{22} & \cdots & \Delta_{2G_i} \\ \vdots & \vdots & \ddots & \vdots \\ \Delta_{G_{i-1}1} & \Delta_{G_{i-1}2} & \cdots & \Delta_{G_{i-1}G_i} \end{bmatrix} \quad (30)$$

Where, G_i is the number of detected line-spectrum located at the i -th region. And the element in the distance matrix is denoted by

$$\Delta_{j,r} = \frac{\sum_{s=1}^{spotnum} \left(w_f |f_{i-1}^j(k_s) - f_i^r(k_s)| + w_{\theta} |\theta_{i-1}^j(k_s) - \theta_i^r(k_s)| \right)}{spotnum} \quad (31)$$

with $1 \leq s \leq spotnum$. It represents the distance between two different line-spectrum. And $spotnum$ is the number of line-spectrum points with the same time information in the two

detected line-spectrum; $f_i^r(k_s)$ and $\theta_i^r(k_s)$ are the frequency and azimuth states of the s -th point in the r -th line-spectrum located at the i -th region. k_s is the time sequence number of the point in line-spectrum. w_f and w_θ are distance weights in frequency and azimuth dimensions respectively.

Two line-spectrum will be merged if their distance $\Delta_{j,r}$ meets the preset threshold conditions. Both azimuth and frequency of line-spectrum points with the same time information in the two line-spectrum are averaged when merging. Fig. 6 shows the process of this step. Then, the line-spectrum merging result is judged again to eliminate the merged line-spectrum whose length does not satisfy the threshold condition. The azimuth and frequency information of the retained line-spectrum is the result of detection.

5. Performance evaluation

In this section, the calculation amount of the proposed method when detecting line-spectrum is firstly analyzed. And then the SFJDT method proposed in this paper is compared with other methods under different SNR conditions. At last, the detection probability curve is given and the processing results of simulation data and actual data are shown in the time sequence of FRAZ spectra.

5.1. Calculation analysis

The amount of computation is too large to meet the requirements of the efficiency of the proposed algorithm under the actual conditions when the dynamic programming method directly processed on the whole FRAZ spectra. Therefore, the method of dynamic searching in different regions is designed to solve the above problem. In Section 2.2, region segmentation can reduce the number of hidden states per iteration, and region screening can eliminate the bearing-time-frequency region where there is no obvious target. The computation of line-spectrum detection is greatly reduced through the above two steps. The following is the analysis of the computation of line-spectrum detection using dynamic programming method before and after region segmentation and screening.

It is assumed that the total number of data segments is K and the number of hidden states is N for each data segment when using Viterbi algorithm to detect line-spectrum. For each iteration, the total number of multiplications is $(5K - 4)N^2 + (K + 1)N$, the total addition number is $(3K - 2)N^2 + (2 - 3K)N$, and the total number of assignments is $3KN$.

The purpose of this paper is to detect line-spectrum in time sequence of the FRAZ spectra. For each segment of data, we hence set the frequency points for beamforming as N , and the number of beam directions is M ; The total number of FRAZ spectra calculated is K ; The preset number of existing line spectrum is L . Before region segmentation, the number of hidden states per segment is MN . In this case, if the line-spectrum detection is carried out directly, the total number of multiplications is

$$Caltr_{Mul} = L \left[(5K - 4)M^2N^2 + (K + 1)MN \right] \quad (32)$$

the total number of additions is

$$Caltr_{Add} = L \left[(3K - 2)M^2N^2 + (2 - 3K)MN \right] \quad (33)$$

the total number of assignments is

$$Caltr_{Assign} = 3LKMN \quad (34)$$

After region segmentation, suppose each sub-region is $N_b * N_f$ rectangular, that is, each region contains N_b beam directions and N_f frequency states; The number of data segments processed each time is set to N_t . The overlap mode is set to overlap 1/2 region on both time dimension and frequency dimension; No overlapping regions are set on the azimuth dimension. Therefore, the number of regions in frequency dimension is $D_1 = 2N/N_f - 1$; the number of regions in azimuth dimension is $D_2 = M/N_b$; the number of regions in time dimension is $D_3 = 2K/N_t - 1$. Since a single target may radiate multiple line-spectrum signals, which are all located in the same direction, the number of possible line-spectrum in the sub-region is set to N_fL/N . Under the above conditions, the number of sub-regions for line-spectrum detection is N . And the number of regions to be detected is $D_1D_2D_3$ if region screening is not carried out. The total number of multiplications required for line-spectrum detection in all sub-regions is

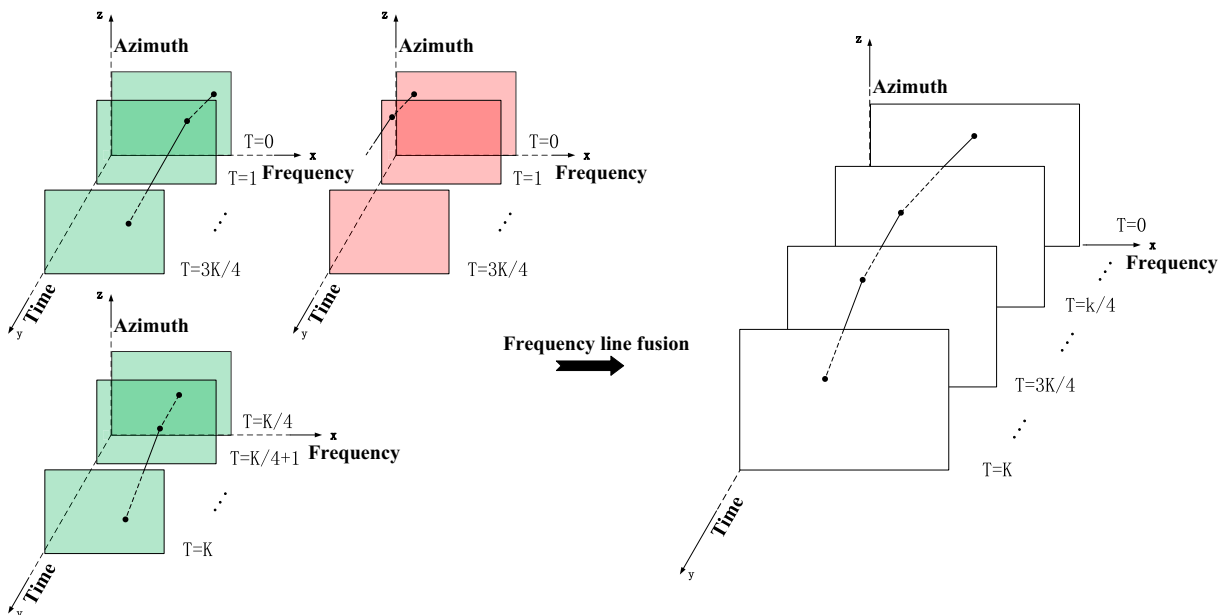


Fig. 6. Schematic showing line-spectrum merging.

$$Cal_{Mul} = \frac{D_1 D_2 D_3 N_f L}{N} \left[(5N_t - 4) N_b^2 N_f^2 + (N_t + 1) N_b N_f \right] \quad (35)$$

the total number of additions is

$$Cal_{Mul} = \frac{D_1 D_2 D_3 N_f L}{N} \left[(3N_t - 2) N_b^2 N_f^2 + (2 - N_t) N_b N_f \right] \quad (36)$$

the total number of assignments is

$$Cal_{tr}_{Assign} = \frac{3D_1 D_2 D_3 N_t N_b N_f^2 L}{N} \quad (37)$$

Replace N_t , N_b , N_f in Eqs (35)–(37) with the expression composed of variables D_1 , D_2 , D_3 , N , M and K . When $D_1 \gg 1$, $D_2 \gg 1$, $D_3 \gg 1$, the approximate calculation amount of line-spectrum detection before and after region segmentation is as shown in Table 1.

As can be seen from the above table, under ideal conditions, the calculation amount after region segmentation is about $D_1^2 D_2 / 16$ times of that before region segmentation. When the size of the whole dataset is determined, the number of regions is inversely proportional to the amount of calculation in line-spectrum detecting. In addition, the above calculation amount analysis is carried out under the condition that the target radiated line-spectrum exists in each sub-region, and all sub regions need to be processed by line-spectrum detection. Moreover, the high false alarm decision made on the sub-regions will eliminate the regions without obvious line-spectrum and further reduce the calculation amount of the algorithm. Accordingly, after region segmentation, region screening and line-spectrum merging will also increase the calculation amount of the algorithm. However, the calculation amount of these two steps is linearly related to the number of regions and the number of line-spectrum respectively, which is far less than that of line-spectrum detection.

Next, the calculation efficiency of the algorithm is tested on the MATLAB simulation platform in PC. In the simulation, linear array is used as the array. The number of effective elements of linear array is $Q = 64$; the distance d between adjacent elements is $1.2m$; the sound velocity $C = 1500m/s$; the number of beamforming directions is 65 and equal cosine segmentation is used; the sampling rate $f_s = 2500Hz$; the number of sampling points used to obtain the power spectra is 4096; the frequency band for frequency domain beamforming is 1–625 Hz; the total number of data segments $K = 100$. Therefore, the observation time span is 163.84 s. Three targets are simulated in the simulation. The angle θ_1 between the first target and the normal direction of the array is 30° and the change rate of target azimuth is $0.02^\circ/s$. The target radiation signal consists of three line-spectrum signals; The angle θ_2 between the second target and the normal direction of the array is 90° . During the observation time, the target azimuth remains unchanged. The target radiation signal consists of three line-spectrum signals; The angle θ_3 between the third target and the normal direction of the array is 150° . The target azimuth remains unchanged. The target radiation signal consists of three line-spectrum signals. Three parameter model is used to simulate the background noise in marine environment. For each sub-region, $N_b = 5$, $N_f = 200$, $N_t = 20$. The overlap mode is set to overlap 1/2 region on both time dimension and frequency dimension; No

overlapping regions are set on the azimuth dimension. The preset number of existing line spectrum $L = 4$. Under the above conditions, it takes 23.295 s to detect all 9 line-spectrum signals by using the algorithm proposed this paper.

It can be concluded that the designed dynamic searching method based on region segmentation and merging similar line-spectrum in this paper can effectively improve the computational efficiency of Viterbi algorithm in line-spectrum detection from the above calculation analysis and simulation results. And the proposed algorithm can realize the real-time detection of line-spectrum on personal PC platform.

5.2. Detection performance comparison

The background equalization in Section 2.1 is usually needed in the actual process of line-spectrum detection, so the background noise is taken as broadband white noise and the target radiation signal is assumed to only have stable line-spectrum in simulation. The center frequency f_1 of line-spectrum signal is 252Hz and the angle θ between the target and the normal direction of the array is 90° ; The number of effective elements of linear array is $Q = 32$; the distance d between adjacent elements is $1.5m$; the sound velocity $C = 1500m/s$; the number of beamforming directions is 180; the sampling rate $f_s = 5000Hz$; the number of sampling points used to obtain the power spectra is 1000; the frequency band for frequency domain beamforming is 100 ~ 400 Hz.

In order to more intuitively compare the performance of different types of algorithms, the probability of detection is defined as follows

$$P_D = \frac{\text{Number of correctly detected frequency points}}{\text{Number of real frequency points}} \times 100\% \quad (38)$$

The detection performance of various methods is then compared in frequency dimension and azimuth dimension.

5.2.1. Frequency dimension

In this part, we compare the proposed method with the traditional line-spectrum detection method and HMM based time-frequency graph detection (TFGD) method [13]. And when using the traditional line-spectrum detection method, we set the rejection threshold based on the OTA (order truncate average) technique [17] to realize the line-spectrum detection. Fig. 7 shows P_D of these three methods at different broad-band SNR.

The detection of line-spectrum using the latter two methods are performed in lofargram, so the azimuth of line-spectrum is known by default when processing. That is to say, the detection is carried out on the beamforming results at the target Azimuth. As can be seen from Fig. 7, the performance of the proposed method in frequency dimension is similar to that of TFGD method, and it is obviously superior to the traditional method. Under the condition of low SNR, it can also effectively extract the frequency information of line-spectrum signal.

5.2.2. Azimuth dimension

In azimuth dimension, this paper also simulates HMM based known line-spectrum detection (HMM-KLSD) method to express the upper performance bound of HMM based azimuth detection method. This method is to determine the azimuth of line-spectrum by processing the azimuth history map composed of the beam output results at the line-spectrum frequency in the same way as in [7] instead of the energy azimuth history map. At different broad-band SNR, the comparison results of the proposed method, FVWD method [5], ASD method [6], HMM based energy detection (HMM-ED) method [7] and HMM-KLSD method are shown in Fig. 8.

Table 1
Comparison of the approximate calculation amount of detecting line-spectrum before and after region segmentation.

Operation	After region segmentation	Before region segmentation
Multiplication	$16L \left(\frac{5KM^2N^2}{D_1^2D_2} + \frac{KMN}{2D_1} \right)$	$L(5KM^2N^2 + KMN)$
Addition	$16L \left(\frac{3KM^2N^2}{D_1^2D_2} - \frac{3KMN}{2D_1} \right)$	$L(3KM^2N^2 - 3KMN)$
Assignment	$\frac{24LKMN}{D_1}$	$3LKMN$

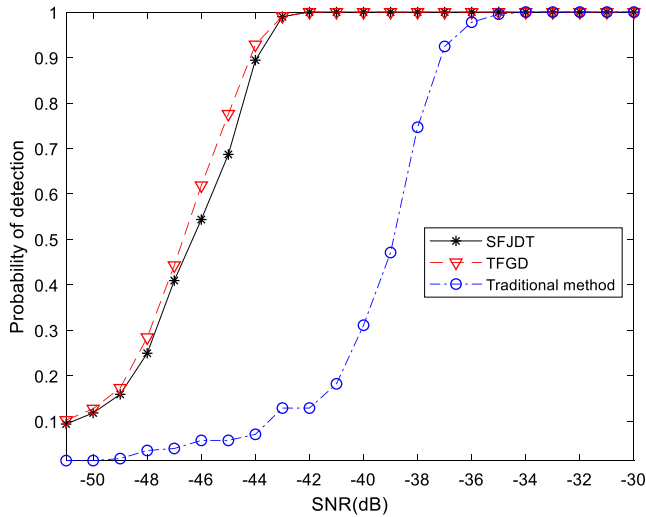


Fig. 7. Performance comparison of different methods in frequency dimension.

It can be concluded from Fig. 8:

- (1) The line-spectrum detection method based on HMM has a good detection performance under the condition of low SNR, and has a higher detection probability than other methods under the same condition.
- (2) The detection performance of the method proposed in this paper is better than the energy detection method based on HMM. And its performance is very close to the known line-spectrum detection method based on HMM.

Combining the results of Sections 4.1.1 and 4.1.2, it can be seen that the line-spectrum signal extraction performance of the proposed method has obvious advantages compared with the traditional method. In the frequency dimension and the azimuth dimension, the target information can be accurately extracted.

5.3. Detecting results of simulation data and actual data

5.3.1. Detecting results of simulation data

In the simulation, linear array is used as the array. The number of effective elements of linear array is $Q = 64$; the distance d between adjacent elements is 1.2m; the sound velocity

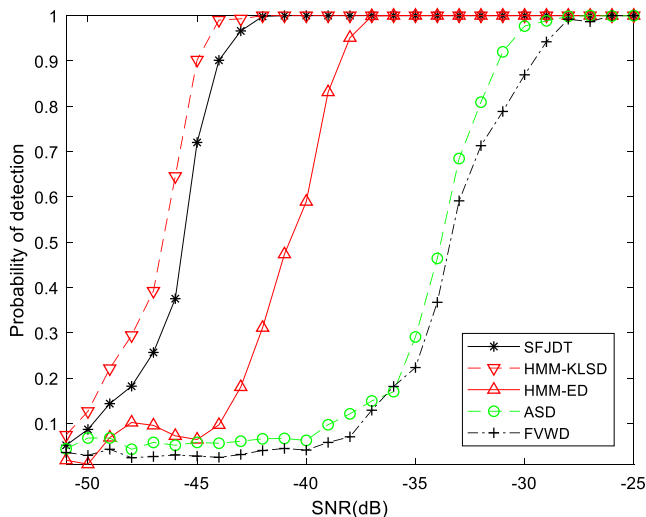


Fig. 8. Performance comparison of different methods in azimuth dimension.

$C = 1500\text{m/s}$; the number of beamforming directions is 65 and equal cosine segmentation is used; the sampling rate $f_s = 2500\text{Hz}$; the number of sampling points used to obtain the power spectra is 4096; the frequency band for frequency domain beamforming is 1–625 Hz; the total number of data segments $K = 100$. Therefore, the observation time span is 163.84 s. Three targets are simulated in the experiment. The angle θ_1 between the first target and the normal direction of the array is 30° and the change rate of target azimuth is $0.03^\circ/\text{s}$. The target radiation signal consists of five line-spectrum signals with frequencies of $f_1 = 41\text{Hz}$, $f_2 = 141\text{Hz}$, $f_3 = 301\text{Hz}$, $f_4 = 401\text{Hz}$, $f_5 = 451\text{Hz}$ and broadband SNR of -30dB , -30dB , -33dB , -34dB , -36dB ; The angle θ_2 between the second target and the normal direction of the array is 90° and the change rate of its azimuth is $0.043^\circ/\text{s}$. The target radiation signal consists of three line-spectrum signals with frequencies of $f_1 = 52\text{Hz}$, $f_2 = 252\text{Hz}$, $f_3 = 412\text{Hz}$ and broadband SNR of -34dB , -40dB , -40dB ; The angle θ_3 between the third target and the normal direction of the array is 150° and the change rate of its azimuth is . The target radiation signal consists of four line-spectrum signals with frequencies of $f_1 = 63\text{Hz}$, $f_2 = 343\text{Hz}$, $f_3 = 433\text{Hz}$, $f_4 = 493\text{Hz}$ and broadband SNR of -30dB , -32dB , -33dB , -35dB . Three parameter model is used to simulate the background noise in marine environment. Under the above conditions, the observation time span is 163.84 s; the azimuth of the first target changes from 30° to 35° ; the azimuth of the first target remains unchanged at 90° ; the azimuth of the first target changes from 150° to 142° .

When detecting line-spectrum using the proposed method, for each sub-region, $N_b = 5$, $N_f = 200$, $N_t = 20$. The overlap mode is set to overlap 1/2 region on both time dimension and frequency dimension; No overlapping regions are set on the azimuth dimension. The preset number of existing line spectrum $L = 4$. And when setting the HMM parameters, we make $\delta_f = 0.8$, $\delta_\theta = 0.5$. And the detection results of the proposed method in the time history of FRAZ spectra are shown in Fig. 9.

It takes 23.512 s to detect all line-spectrum signals on the MATLAB simulation platform in PC using SFJDT method. And the azimuth and frequency information of the detected line-spectrum signals are compared with the actual information in Table 2. In the fourth and fifth columns of the table, the expression $\theta_1 \rightarrow \theta_2$ means the azimuth of the target changes from θ_1 to θ_2 . It can be seen from the table that all the 12 line-spectrum signals in the simulation data are extracted. And the azimuth and frequency information of each line-spectrum is basically correct. Although there exists some deviation in azimuth and frequency dimensions, it is influenced by the number of array elements and frequency resolution, and has little to do with the proposed algorithm.

Besides, the traditional detection before track (DBT) method are used to process the above simulation data in this part for performance comparison. When DBT is used for detection, the first step is to obtain the frequency and azimuth information of the points in each FRAZ spectra that meet the threshold conditions by threshold detection. Nearest neighbor data association is then performed on the detection results of multi-frame FRAZ spectra to obtain the complete line-spectrum information. And the detection results of the two methods using the same simulation data are shown in Fig. 9.

It can be seen from the Fig. 9 that when detecting the line-spectrum generated by target 1 and target 3, the detection results of the proposed method have less fluctuation in the azimuth dimension compared with that of the traditional DBT method. It means that the SFJDT method can generate less false alarms in azimuth dimension under the condition of high SNR. And when detecting the line-spectrum generated by target 2,

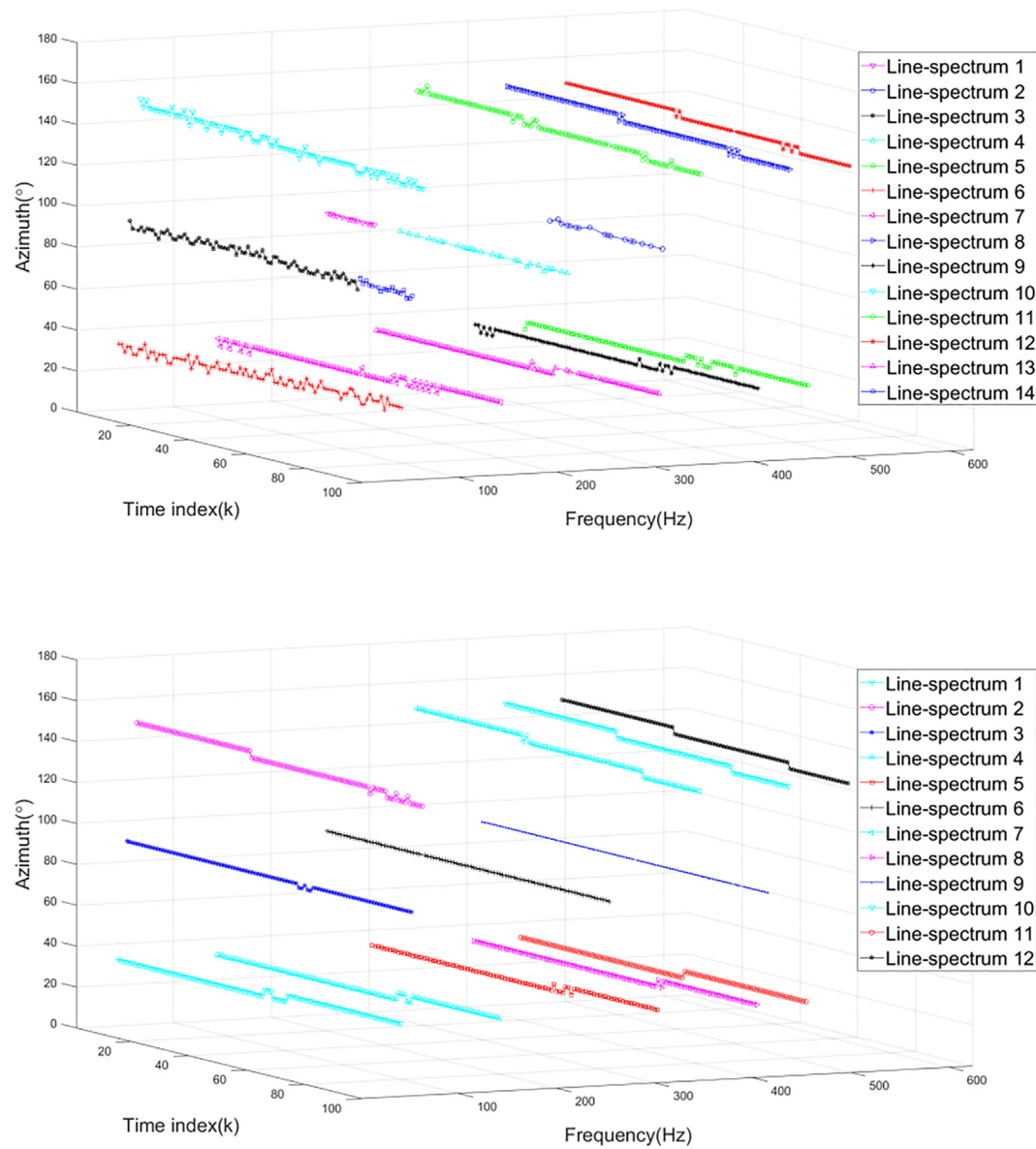


Fig. 9. (Top) Detection results by DBT method. (Bottom) Detection results by the proposed SFJDT method.

Table 2

Comparison of the azimuth and frequency information of actual signals and detected line-spectrum signals.

Target	Actual frequency	Detected frequency	Actual azimuth	Detected azimuth
1	41 Hz	40.89 Hz	30°→35°	32°→35.6°
	141 Hz	141.6 Hz	30°→35°	32°→35.6°
	301 Hz	303.3 Hz	30°→35°	32°→35.6°
	401 Hz	404.1 Hz	30°→35°	32°→35.6°
	451 Hz	454.7 Hz	30°→35°	32°→35.6°
2	52 Hz	51.88 Hz	90°	90°
	252 Hz	253.9 Hz	90°	90°
	412 Hz	415.6 Hz	90°	90°
3	63 Hz	62.87 Hz	150°→142°	147.5°→141.4°
	343 Hz	345.5 Hz	150°→142°	147.5°→141.4°
	433 Hz	436.4 Hz	150°→142°	147.5°→141.4°
	493 Hz	497.4 Hz	150°→142°	147.5°→141.4°

the line spectrum detected by traditional methods not only has greater fluctuation, but also is incomplete (there are breakpoints in some line-spectrum). It means that under the condition of low SNR, the proposed method can not only reduce the false

alarm in azimuth dimension, but also has a higher detection probability. Therefore, we can conclude that the proposed method has better performance compared with traditional DBT method.

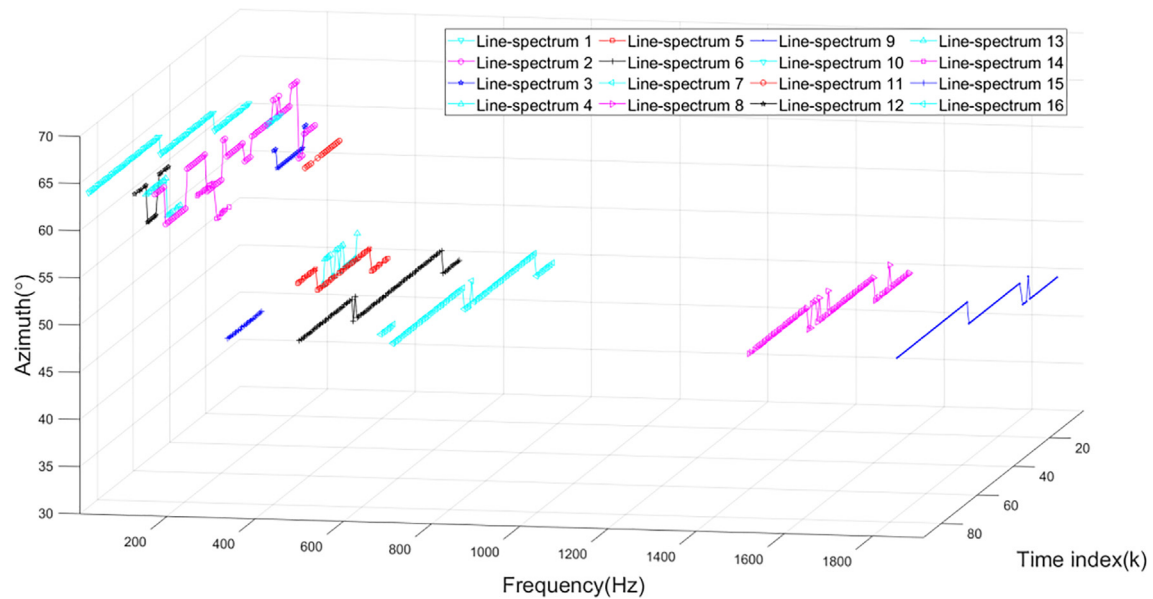


Fig. 10. Detection results of actual data in the time history of FRAZ spectra.

5.3.2. Detecting results of actual data

We also use this method to process the actual data. The actual data is obtained at sea by a linear array sonar towed at the stern of a ship. The sonar array is composed of five sub-arrays SA1–SA5 with different array element spacing. The operating frequency range is 20–1875 Hz. The number of effective elements of SA1–SA3 is 32, and that of SA4–SA5 is 50. The sampling rate is 32 kHz. During the period of data acquisition, there were two ships sailing on the water surface at more than ten kilometers, respectively in the direction of angle $\theta_1 \approx 42^\circ$ and $\theta_2 \approx 60^\circ$.

Fig. 10 shows the line-spectrum detection results of actual data in the time history of FRAZ spectra by the proposed method. The observation time span of actual data is about 80 s. It takes 20.399 s to detect all line-spectrum signals on the MATLAB simulation platform in PC. And the proposed algorithm in this paper has detected a total of 16 line-spectrum signals from the actual data, which are concentrated in the 44.05° – 48.99° and 60° – 64.06° directions of the linear array. The results are consistent with the actual data and show that this method is still effective for actual data.

It can be concluded from Sections 4.3.1 and 4.3.2 that the SFJDT method can effectively detect the target and accurately extract the frequency and azimuth information of the target-radiated line-spectrum signal. And the method has better detection performance compared with traditional methods under the situation when there are multiple line-spectrum signals in different azimuth.

6. Conclusions

The detection of the azimuth and frequency information of the target radiated line-spectrum signal under the condition of low SNR is studied in this paper. A space-frequency joint detection and tracking method for line-spectrum signals based on TBD is proposed. In the paper, the change of frequency and azimuth of the target radiated line-spectrum signal is modeled as a hidden Markov process, and the parameters of HMM are derived theoretically according to the distribution of the FRAZ spectra. Combined with Viterbi algorithm and the idea of data association, the simultaneous extraction and fusion of azimuth and frequency information of line-spectrum signal are realized. A dynamic searching method based on region segmentation and merging similar line-

spectrum is designed to solve the problem of computation in the extraction of line-spectrum signals, which makes the whole detection process meet the requirements of real-time processing. By combining TBD framework with HMM between blocks, Viterbi algorithm and data association, the correlation of the target information in multi-frame FRAZ spectra is fully utilized to improve the performance of line-spectrum detection. It can be concluded from the processing results of simulation data and actual data that the proposed method has better detection performance and requires less prior information under the condition of low SNR compared with other methods. The method can adapt to the complex situation when there are multiple time-varying line-spectrum signals in different azimuth. The proposed SFJDT method has theoretical value and practical significance for improving the detection ability of unknown line-spectrum target in complex underwater acoustic environment.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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